A formal homogeneous Jacobian ideal separator

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Abstract

We formalize an explicit second-order differential operator outside the full algebra generated by multiplication operators and all derivations for every ordinary homogeneous polynomial of degree at least two in at least three variables over a characteristic-zero field, assuming its gradient quotient is finite-dimensional. An elementary determinant comparison with pure powers proves the required Hessian nonmembership inside Lean, removing the classical socle theorem previously left as an external mathematical input. The complete sixteen-module proof and an expanded specification pass an independent fresh-output rebuild. This extends the formal scope of the fixed-quartic certificate; the operator and Jacobian-ideal stability retain their prior attributions. Mathematical priority is not established. The weighted case and the unrestricted Nakai conjecture are not resolved here.

1 Exact scope and operator

Let k be a field of characteristic zero, let $P = k[x_1, \dots, x_n]$, and write

$$f_i = \partial_i f, \quad J = (f_1, \dots, f_n), \quad H = (\partial_i \partial_j f)_{ij}, \quad U = \operatorname{adj}(H), \quad h = \det H.$$

For a k-algebra A, put

$$\Delta_k(A) = k\langle m_a, \delta : a \in A,\ \delta \in \operatorname{Der}_k(A)\rangle \subseteq \operatorname{End}_k(A), \qquad m_a(b) = ab.$$

There is no bound on the number of derivation factors in this algebra. Differential order at most two means

$$[[[T, m_a], m_b], m_c] = 0 \qquad (a, b, c \in A). \tag{1}$$

This is the usual commutator characterization on actual endomorphisms.

Theorem 1.1. *Suppose $n \geq 3$, $d \geq 2$, f is homogeneous of degree d, and $\dim_k(P/J) < \infty$. Fix $r \in \{1, \dots, n\}$. Define*

$$B_{ij} = U_{ir} x_j + x_i U_{jr} - x_r U_{ij}, \tag{2}$$

$$Q_r(p) = \sum_{i,j} B_{ij} \partial_i \partial_j p + (n-2)(d-1) \sum_i U_{ir} \partial_i p. \tag{3}$$

Then Q_r descends to an actual k-linear endomorphism $\overline{Q}_r$ of $A = P/(f)$, satisfies (1)*, and $\overline{Q}_r \notin \Delta_k(A)$.*

The quotient finiteness hypothesis is expressed as `Module.Finite` over the field. No Hessian nonmembership, regular-sequence, or socle assertion is assumed. The algebraic operator theorem is formalized for arbitrary finite index types. Geometric translations of the finiteness hypothesis are not additional formal statements of this package.

Prior work. The operator (3) is twice the specified ordinary-homogeneous lift in Yau–Zhu–Zuo [1, eqs. (3.1), (3.3), (5.1), (5.6), Cor. 5.5], including its first-order term. In three variables, the corresponding operator of Diethorn et al. is $Q_r/(d-1)^2$ [2, Sections 4.1.3 and 4.5, eqs. (33), (35), Theorem 4.5.1]; older three-variable full-ring results also predate this work. Jacobian-ideal stability is recorded in Li–Xiao–Zuo [3, Theorem 2.7]. Their Remark 1.1 explicitly distinguishes bounded-length generation from intersection with the full generated algebra. The introduction and main theorem of [1] formulate an order-by-order second-order obstruction; we do not use that obstruction alone to infer unrestricted-word separation. The purpose here is a complete checked proof of the stated algebraic theorem, not a priority claim for its ingredients or assembled conclusion.

2 An elementary Hessian nonmembership proof

This section supplies the step that was previously imported from classical complete-intersection socle theory. Determinantal nonmembership and transformation methods are classical; see, for example, Simon–Strooker [4, Propositions 1.2 and 2.1] and Eisenbud–Ulrich [5, Theorem 7.1]. The proof below is chosen to fit a small, explicit formal dependency graph.

Lemma 2.1. *Let M be a square matrix over a commutative ring, x a column vector, and I an ideal containing every entry of Mx. For all r, p, q,*

$$x_q \operatorname{adj}(M)_{pr} - x_p \operatorname{adj}(M)_{qr} \in I.$$

Proof. Replace row r of M by the basis row e_p^T, obtaining L. Then $\det L = \operatorname{adj}(M)_{pr}$, $(Lx)_r = x_p$, and $\operatorname{adj}(L)_{qr} = \operatorname{adj}(M)_{qr}$, because the latter cofactor deletes the replaced row. The q-th entry of $\operatorname{adj}(L)Lx = \det(L)x$ gives

$$x_q \operatorname{adj}(M)_{pr} - x_p \operatorname{adj}(M)_{qr} = \sum_{\ell \neq r} \operatorname{adj}(L)_{q\ell}(Mx)_\ell \in I.$$

□

Lemma 2.2. *Suppose $C, D \in M_n(P)$, every entry of both matrices is homogeneous of the same degree m, and $Cx = Dx = y$. Then $\det C - \det D \in (y_1, \ldots, y_n)$.*

Proof. If $u = (u_j)$ is a row with entries homogeneous of degree m and $\sum_j u_j x_j = 0$, Euler's identity and the derivative of this relation give, for any coefficients c_j,

$$(m+1)\sum_j c_j u_j = \sum_{i,j} (\partial_i u_j)(x_i c_j - x_j c_i). \tag{4}$$

Indeed, the coefficient of c_j on the right is $\sum_i x_i \partial_i u_j - \sum_i x_i \partial_j u_i = m u_j - (-u_j)$. For a matrix M with $Mx = y$, take $c_j = \operatorname{adj}(M)_{jr}$. Every parenthesized term in (4) belongs to (y) by Lemma 2.1. Since $m+1$ is invertible in k, the row determinant $\det(M[r \leftarrow u]) = \sum_j \operatorname{adj}(M)_{jr} u_j$ belongs to (y). Replace the rows of D successively by those of C. Each row difference is such a homogeneous syzygy, and every intermediate matrix still has product y with x. Row linearity and a finite sum of these determinant differences prove the claim. □

Proposition 2.3. *Let $g_1, \ldots, g_n$ be homogeneous of the same positive degree s, let $J = (g_1, \ldots, g_n)$, and suppose P/J is finite-dimensional. If $g = Ex$, where the entries of E are homogeneous of degree $s-1$, then $\det E \notin J$.*

Proof. Each class $[x_i] \in P/J$ is integral over k. Choose a monic polynomial $p_i(t)$ annihilating it. Then $p_i(x_i) \in J$. The ideal J is homogeneous, and the top homogeneous component of $p_i(x_i)$ is $x_i^{\deg p_i}$; hence some power of each variable lies in J. Choose a common exponent $N \geq s$, $N > 0$, at least as large as all those exponents. Thus $x_i^N = \sum_j b_{ij} g_j$.

Taking degree N components replaces every b_{ij} by its component of degree $N - s$, without changing this identity. The fact used here is

$$[bg]_N = [b]_{N-s} g \quad \text{if } g \text{ is homogeneous of degree } s \text{ and } N \geq s.$$

Let $B = (b_{ij})$ denote these homogeneous coefficients. Then $BEx = (x_i^N)_i$. The entries of BE and of $D = \operatorname{diag}(x_i^{N-1})$ have common degree $N - 1$. Lemma 2.2, with $I = (x_1^N, \ldots, x_n^N)$, gives

$$\det B \det E - \prod_i x_i^{N-1} \in I. \tag{5}$$

Also $Bg = (x_i^N)_i$, so the adjugate identity gives $\det B \, g = \operatorname{adj}(B)(x_i^N)_i$. Consequently $\det B \, J \subseteq I$. If $\det E \in J$, equation (5) would imply $\prod_i x_i^{N-1} \in I$. This is impossible: none of the generators x_i^N divides that monomial. The monomial-ideal membership criterion is checked for the actual polynomial ideal in the formal proof. □

Apply Proposition 2.3 to $g_i = f_i$, $s = d - 1$, and $E = (d-1)^{-1} H$. Differentiated Euler gives $Hx = (d-1)\nabla f$, so $Ex = \nabla f$. The proposition yields

$$h = \det H \notin J. \tag{6}$$

Thus the full Hessian nonmembership conclusion, rather than a classical socle theorem used as an assumption, is now a kernel-checked consequence of finiteness and homogeneity.

3 Descent and unrestricted word separation

The generic operator modules establish the following exact polynomial identities:

$$Q_r(fp) = f\left(Q_r(p) + 2d \sum_i U_{ir} \partial_i p\right) \qquad (p \in P), \tag{7}$$

$$Q_r(f_r) = (nd - n - d)h - (d-1) \sum_i (\partial_r U_{ir}) f_i. \tag{8}$$

Equation (7) proves descent on every representative. The expression (3) is a sum of polynomial multiples of compositions of at most two partial derivations; the formal commutator proof gives (1) on P and transports it to $P/(f)$. The exact first-order term is included throughout.

Euler's identity gives $f \in J$. Every derivation δ of A can be lifted, by lifting its values on the variables, to $\widetilde{\delta} = \sum_i u_i \partial_i$ on P, with $\widetilde{\delta}(f) = vf$. Differentiating gives

$$\widetilde{\delta}(f_j) = (\partial_j v) f + v f_j - \sum_i (\partial_j u_i) f_i \in J.$$

Hence every actual quotient derivation preserves the image $\overline{J}$ of J in A. Multiplications preserve it too, so the entire subalgebra $\Delta_k(A)$ preserves $\overline{J}$. This includes arbitrary finite words and all cancellations among them; no classification or selected list of derivation generators is used.

Since

$$nd - n - d = (n-2)(d-1) + (d-2) > 0 \quad (n \geq 3,\ d \geq 2),$$

this integer is nonzero in k. Equations (6) and (8) imply $Q_r(f_r) \notin J$. Because $(f) \subseteq J$, its quotient class is outside $\overline{J}$. Thus $\overline{Q}_r$ fails to preserve an ideal preserved by every element of $\Delta_k(A)$, proving Theorem 1.1.

4 Formal verification and limits

The checked final theorem is

`NakaiHomogeneousFiniteExpected.`
`expected_finite_gradient_full_operator_separation`.

Its expanded statement writes the polynomial operator, literal gradient quotient, actual hypersurface quotient, triple commutators, and unrestricted `Algebra.adjoin` of all multiplication maps and all quotient derivations. The only geometric-algebraic input is `Module.Finite` of the gradient quotient. The determinant and finite-quotient lemmas prove the missing nonmembership implication, without assuming a socle result.

All sixteen project modules were independently rebuilt into a fresh output directory with Lean 4.32.2 and the following exact pins:

Lean: `f3b06c705e6c85f5314019d5d3baab0fec5b580c`
Mathlib: `905b95818eb32af7874a58b427f50c1711a5e96c`.

One unchanged pinned upstream module, `MvPolynomial/EulerIdentity`, was compiled privately using already cached dependencies. No previous project object files were used. Every module exited successfully; the final axiom report lists only `propext`, `Classical.choice`, and `Quot.sound`. Exact source hashes, logs, guards, and the reproducible dependency order are included with the source package.

This addendum supersedes the earlier fixed-quartic note's limitation that the general ordinary-homogeneous statement was only a written theorem. It does not alter that note's fifteen-source concrete certificate or its operator attribution. It does not formalize a weighted-homogeneous extension, settle the general nonhomogeneous case, or establish historical novelty of the full-word argument. Those distinctions remain explicit.

References

[1] S. S.-T. Yau, Q. Zhu, and H. Zuo, *Nakai conjecture for isolated weighted homogeneous hypersurface singularities*, arXiv:2604.24508v4, 8 October 2026. `https://arxiv.org/html/2604.24508v4`.

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[3] R. Li, Z. Xiao, and H. Zuo, *The Nakai Conjecture for isolated hypersurface singularities of modality* ≤ 2, arXiv:2502.04672v1. `https://arxiv.org/pdf/2502.04672v1`.

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[5] D. Eisenbud and B. Ulrich, *Duality and socle generators for residual intersections*, Journal für die reine und angewandte Mathematik **756** (2019), 183–226. `https://par.nsf.gov/servlets/purl/10282767`.